

1



DATOS

$$\begin{array}{l} m_v = 120g \\ m_{v+H_2O} = 190g \end{array} \left\{ \begin{array}{l} m_{H_2O} = 70g \\ \rho_{H_2O} = 1g/ml \end{array} \right\} \rightarrow V_{H_2O} = \frac{m}{\rho} = \frac{70g}{1g/ml} = 70ml$$

P/METAL: $m_M = 10g$

$$\begin{array}{l} m_{v+n} = 120 + 10 = 130g \\ m_{v+n} + m_{H_2O} = 194g \end{array} \left\{ \begin{array}{l} m_{H_2O} = 64g \\ V_{H_2O} = 64ml \end{array} \right.$$

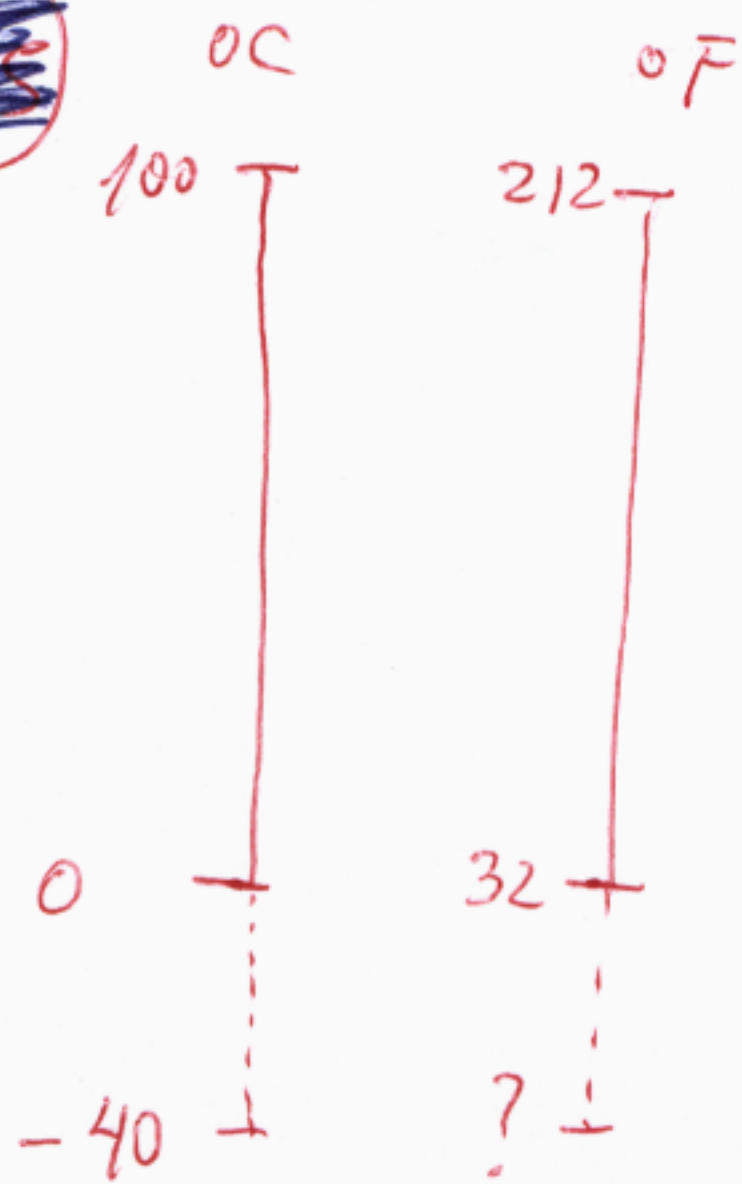
METAL OCUPA UN VOLUMEN:

$$70ml - 64ml = 6ml$$

$$\therefore \rho_M = \frac{m_M}{V_M} = \frac{10g}{6ml} = 1.67g/ml$$

R: (C)

2



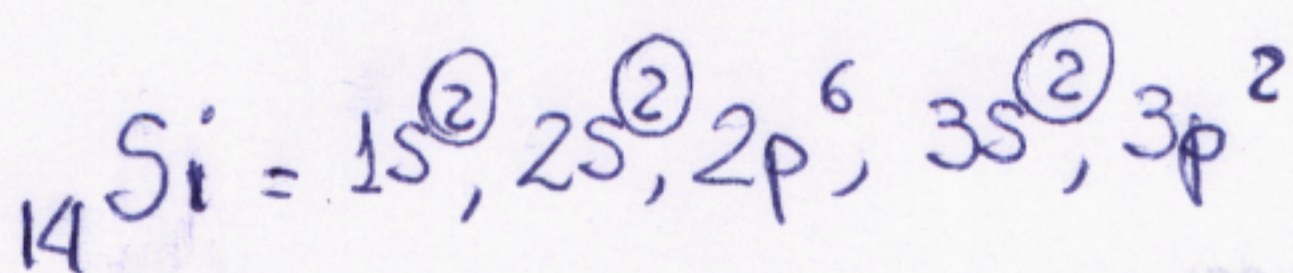
$$\frac{OF - 32}{-40 - 0} = \frac{212 - 32}{100 - 0}$$

$$OF = \frac{9}{5}(-40) + 32 = -72 + 32$$

$$OF = -40$$

R. - (D)

ESTRUCTURA ATÓMICA



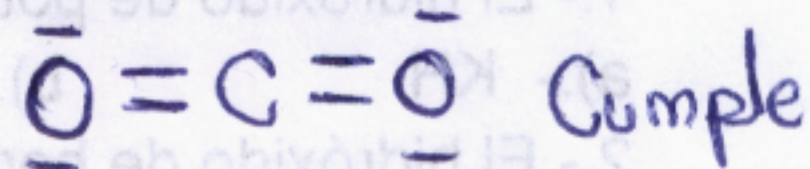
Número total de electrones "S" = $2+2+2 = 6e^-$

Resp. **D) 6**

ENLACE QUÍMICO

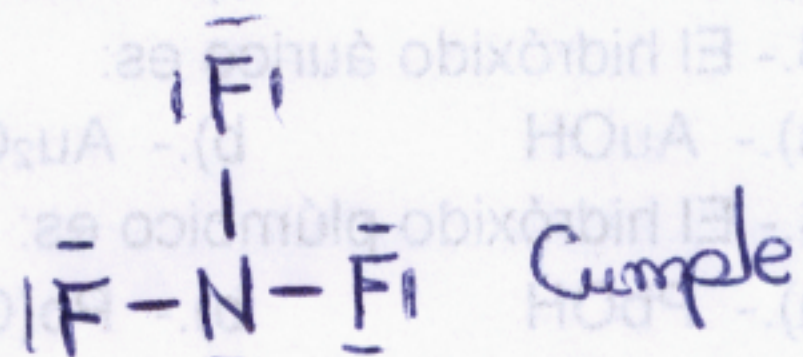
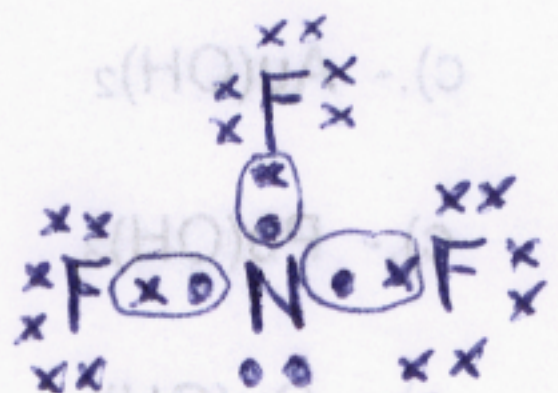
④ a) CO_2

$\Delta_{\text{C-O}} = 3,5 - 2,5 = 1 \text{ (E. Covalente)}$



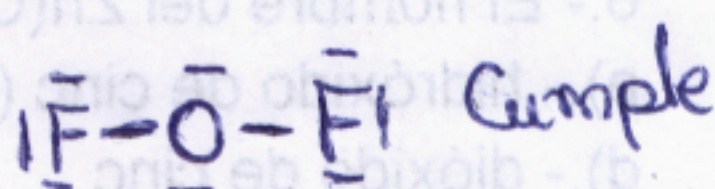
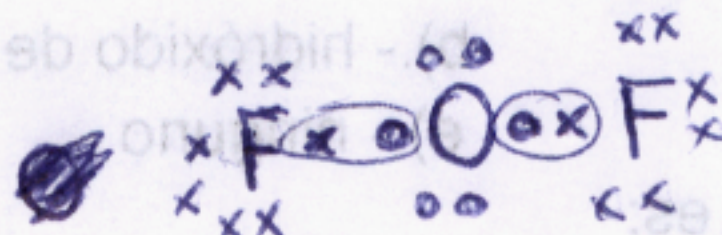
b) NF_3

$\Delta_{\text{N-F}} = 4,1 - 3,1 = 1 \text{ (E. Covalente)}$



c) OF_2

$\Delta_{\text{O-F}} = 4,1 - 3,5 = 0,6 \text{ (E. Covalente)}$



d) PF_5

$\Delta_{\text{P-F}} = 4,1 - 2,1 = 2 \text{ (E. Iónico)}$



No Cumple con la Regla del Octeto

Resp. **D) PF_5**

FUNDAMENTOS DE ESTEQUIOMETRÍA

5

4 at. Fe

$6,023 \times 10^{23}$ moles hemog.

1 mol Fe

56 g Fe

1 moles hemog.

1 mol hemog.

~~6,023~~ $6,023 \times 10^{23}$ at Fe

1 mol Fe

$$\times \frac{100 \text{ g hemog.}}{0,34 \text{ g Fe}} = 65882,35 \text{ g/mol}$$

Resp.

c) 65882,4